

Kaivalya Mase

B.Tech Mechanical Engineering

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Education

Degree	Institute	Year	CGPA/%
B.Tech Mechanical Engineering	IIT Hyderabad	2027	9.4
B.Tech CSE (Double Major)	IIT Hyderabad	2027	6.5
Class XII	Late PB Jog Junior College	2023	89.83%

Internships

Applied Materials India

May 2026 – Jul 2026

Bengaluru, India

Summer Intern - Atomic Layer Deposition

- Underwent semiconductor manufacturing and chip fabrication orientation.
- Problem statement: Extending Preventive Maintenance interval for ALD chamber components.
- Proposed an experimental methodology for quantifying the effect of substrate condition on coating adhesion.

Projects

Formula SAE — Chassis Team

May 2023 – Apr 2026

Supra SAE India 2025

- Designed the space-frame chassis using SolidWorks.
- Performed FEA to identify weak regions and reduce front-impact deformation to approximately 4 mm and a torsional stiffness of approximately 1150 Nm/deg.
- Manufactured the chassis using MIG welding and assembled the suspension.
- Coordinated with Vehicle Dynamics and Powertrain teams for subsystem integration.

Fracture Mechanics Course Project

Apr 2026

- Automated Abaqus simulations for Disc-Shaped Compact Tension specimens.
- Generated geometry, assigned material properties and crack parameters.
- Generated meshes using Barsoum quarter-point elements.
- Extracted stress intensity factors from approximately 800 simulations.

Inter IIT Tech Meet 2025

Oct 2025 – Dec 2025

- Designed and simulated a high-lift airfoil and wing configuration.
- Developed preprocessing workflows in ANSYS Fluent.
- Conducted mesh convergence studies.
- Validated the configuration achieving $C_L \approx 6.5$ during takeoff with target lift-to-drag ratios $L/D \approx 20$ at cruise.
- Secured 4th place among 23 IITs in the problem statement.

Technical Skills

CAD & Modelling: SolidEdge, Fusion360, SolidWorks, MSC Adams

Simulation: ANSYS, Abaqus

Programming: C, Python, MATLAB

Documentation: LaTeX

Relevant Coursework

Mechanics	Engineering Mechanics, Solid Mechanics, Fracture Mechanics, Composite Structures
Thermal	Thermodynamics, Heat Transfer, Fluid Mechanics
Manufacturing	Manufacturing Science, Additive Manufacturing
Design	Theory of Machinery

Certifications

Institute of Cost Accountants of India (ICMAI)

2024

Cleared the Foundation level of the Cost and Management Accountant Examination.